Preventing host discovery and port-scanning

Host discovery uses **`ICMP ping echo`** message

➡ ICMP can be disabled or reserved to hosts on the same network

Port Scanning uses **`TCP-syn`** messages

➡ TCP connections can be rejected if a source attempts to initiate multiple connections on multiple ports simultaneously

➡ **Packet filtering** can prevent these two scanning techniques

Limitation of a host-by-host packet filtering solution

How to enable packet filtering on every host on the network?

1. Each host needs to have **packet filtering capability** across different hardware, OS and versions
2. The admin needs to have **administrative privilege on every host** to push the packet filtering policy

➡ Impossible in practice